

Yaoliang Bian

+86 183-2637-2810 | bianyaoliang@mail.ustc.edu.cn | jeffyaoliang.github.io
Hometown: Lu'an, Anhui, China

Research Interests

Embodied AI and multimodal foundation models, with a focus on (1) cross-modal alignment and projector mechanisms in vision-language models; (2) distributed training and fine-tuning of VLA models; and (3) long-horizon memory construction and retrieval for embodied agents. Hands-on experience with multi-GPU distributed training (DDP), LoRA fine-tuning, and inference deployment (vLLM).

Education

University of Science and Technology of China (USTC)

Sep 2022 – Jun 2026 (expected)

B.Eng. in Electrical Engineering

Selected coursework: Robotics Design and Practice (A+), Electronic Design Practice (A–), Electronics Lab, Computer Programming A, Linear Algebra, Mathematical Analysis.

Honors and Awards

USTC Sunshine Scholarship (1/49)	2025
McMaster–USTC Summer Research Internship Program — Completion Certificate	2024
USTC Fellowship Undergraduate A-Class Funding (Top 5%)	2023
Robogame Competition at USTC, 2nd Prize (2/39)	2023
USTC Sunshine Scholarship (1/45); Scholarship for Electrical Engineering Program of Excellence (for talented students); “Star of Diligent Learning”	2023

Research Highlights

Norm Cliff in VLA Finetuning: When MLLM Calibration Recipes Don't Transfer

Mar 2026 – Present

- **Distributed training stack.** Built a 4-GPU PyTorch DDP pipeline for LoRA fine-tuning of OpenVLA-7B (200K steps, batch=16, BF16 mixed precision, gradient checkpointing), with full optimizer/scheduler/RNG checkpoint resume.
- **Negative finding: norm cliff in VLA action finetuning.** Ran a controlled sweep of 12 projector normalization variants on OpenVLA-7B + LIBERO-Spatial. MLLM calibration recipes that align vision-token norms to the text scale ($T \approx 0.83$, $T \approx 1.08$; Qi 2025 / Li 2025) yield SR **64.8–71.4%**, while baselines preserving the original magnitudes ($T \in [8, 128]$) form a stable plateau at **77–82%** — a $16\times$ T-range plateau and a >10 pp gap.
- **Causal mechanism: γ -rescue intervention.** After 200K-step training, froze the LoRA weights and trained only the projector gain γ for 50K more steps; γ drifted away from the alignment target while SR degraded 5.6 pp, providing causal evidence that projector training locks in at initialization rather than converging toward calibration.
- **Reproducibility anchor.** Pipeline reproduces the official OpenVLA–LIBERO-Spatial checkpoint within 0.6 pp (79.8% vs. 80.4%), confirming setup validity.
- **Output:** Manuscript in preparation (targeting CoRL 2026, May 28 deadline).

Industry Experience

Knowin AI — Algorithm Intern

Dec 2025 – Apr 2026

Advisor: [Prof. Yingcong Chen](#) (Information Hub, HKUST(GZ)).

- **M3-Agent reproduction and deployment** (ICLR 2026, ByteDance Seed). The system builds a persistent memory graph from streaming video by encoding visual, auditory, and spatial signals into a unified graph, and answers long-horizon questions through multi-turn retrieval. Reached **30.9% accuracy on M3-Bench-robot** (1,276 QA, 100 indoor scenes), matching the paper's 30.7%, while cutting per-query latency from **30s to 4s (7 $\times$ speedup)** via 50+ controlled experiments on retrieval parameters, embedding models, and batched inference.
- **Speaker-identity pipeline diagnosis.** Found a three-stage cascading failure: (1) SpeakerLab clustering fragmented 3 speakers into 47 IDs; (2) VLM equivalence judgement produced false positives; (3) Union-Find transitive merging collapsed all IDs into one identity. Replaced the speech model with **CAM++ (3D-Speaker)** embeddings and adaptive-k spectral clustering, reducing per-scene speaker fragments from **25.4 to 5.4** on average.

- **Closed-loop fault-recovery framework with episodic memory.** Designed a Detect → Reflect → Correct → Memorize → Pre-check loop that synthesizes ideas from REFLECT (CoRL 2024), Inner Monologue (CoRL 2022), and Reflexion (NeurIPS 2023); the memory and pre-check components extend prior work.
- **Long-term memory retrieval for LLM agents.** Proposed an embedding-free retrieval scheme based on LLM query rewriting + BM25; matches the embedding baseline overall on the LoCoMo benchmark and **outperforms it by 18.4%** on temporal-reasoning subsets. Provided a theoretical analysis of why embeddings struggle with multi-hop temporal reasoning.
- **Output:** Multiple manuscripts in preparation.

Fiveages Ltd — VLA Algorithm Intern

Sep 2025 – Dec 2025

Advisor: *Prof. Yan Huang* (NLP, Institute of Automation, Chinese Academy of Sciences).

- Worked on VLA training optimization and vision-backbone replacement. On top of **BridgeVLA**, processed and integrated the **VastTrack** dataset for pre-training, then ran distributed re-training to improve cross-scene generalization.
- Replaced the fine-tuning-stage **PaLiGemma** vision backbone with the **SANSA** vision network to strengthen object segmentation and visual grounding. Surveyed VLA + RL literature and explored RL–VLM combinations.

Selected Projects

USTC Robogame 2023

May – Oct 2023

Built an autonomous robot that recognizes and grasps minerals of different colors and delivers them to designated zones on schedule. Used AprilTag for real-time localization, and applied bilateral filtering, edge detection, and HSV-space color segmentation for ore classification. **Second Prize among 39 teams.**

AI Final-Exam Review Agent — Nankesong Hackathon

Jun 28, 2025

Developed a Gemini 2.5 Flash–based agent that lets students upload course material and past exams to build a personal knowledge base, then generates predicted exam questions linked back to the relevant knowledge points. Built both backend and frontend UI. **Won the top “Star of Nanke” prize and 1st place at WaytoAGI Nanjing.**

Custom Hardware for Deep Reinforcement Learning

Jul – Sep 2024

Advisor: *Prof. Ameer Abdelhadi* (ECE, McMaster University). Surveyed GPU and FPGA implementations of DQN, PPO, A3C, and SAC; compared parallelization strategies and bottlenecks across hardware architectures. Authored a comprehensive survey of RL hardware acceleration and proposed near-memory computation + advanced layer fusion as future directions.

Competitions

18th National Undergraduate Smart Car Competition — Anhui Region, Power-Relay Group

Nov 2022 – Jul 2023

Team “Shrimphhead Conan”. Built a self-powered rescue car that traverses slopes, turns, intersections, roundabouts, and roadblocks via sensors and onboard control to recharge a stranded car. *Personal contribution:* PCB design of the core circuit boards for both rescue and stranded cars.

International Exchange

National University of Singapore — Dexterous Manipulation Research

Jul – Sep 2025

University of Texas at Austin — Software Engineering Winter School

Winter 2024

University of Oxford, Oriel College — Quantum Computing Summer School

Summer 2023

University of Warsaw — Information Technology Winter School

Winter 2023

Skills

Programming: Python, PyTorch, C, C++, Java, Verilog

Training: PyTorch DDP, LoRA / PEFT, BF16 mixed precision, gradient checkpointing, vLLM inference deployment

Systems: Git, Docker, SLURM cluster scheduling, GCP (4×A100), Linux

Hardware: FPGA, STM32, Raspberry Pi, nRF52840 SoC

English: CET-6 568 · CET-4 597 · TOEFL 85